



BITS Pilani

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CE F231 Fluid Mechanics

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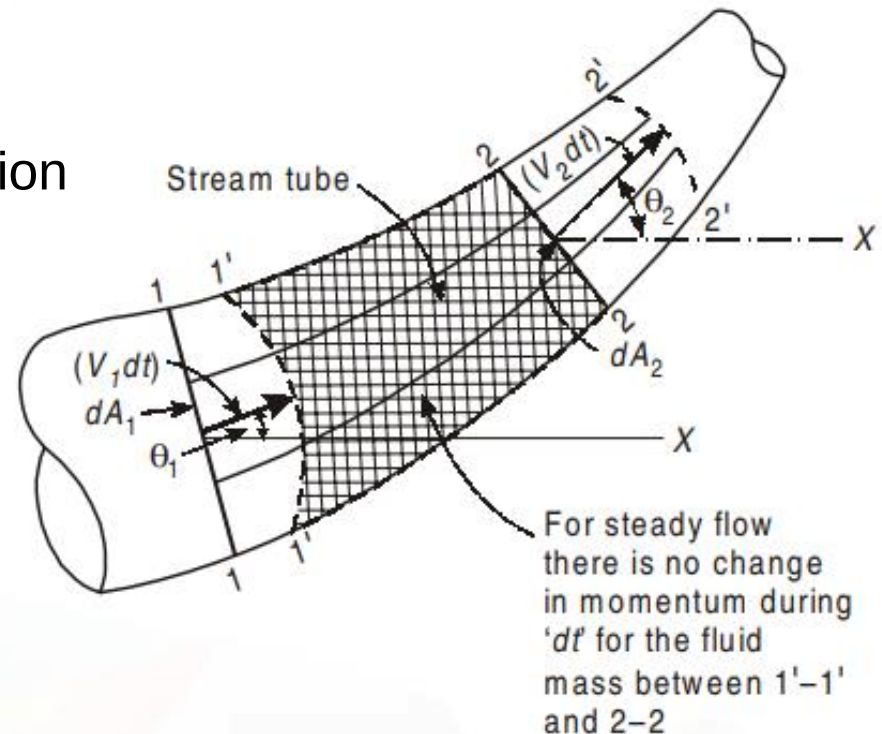
November 14, 2022



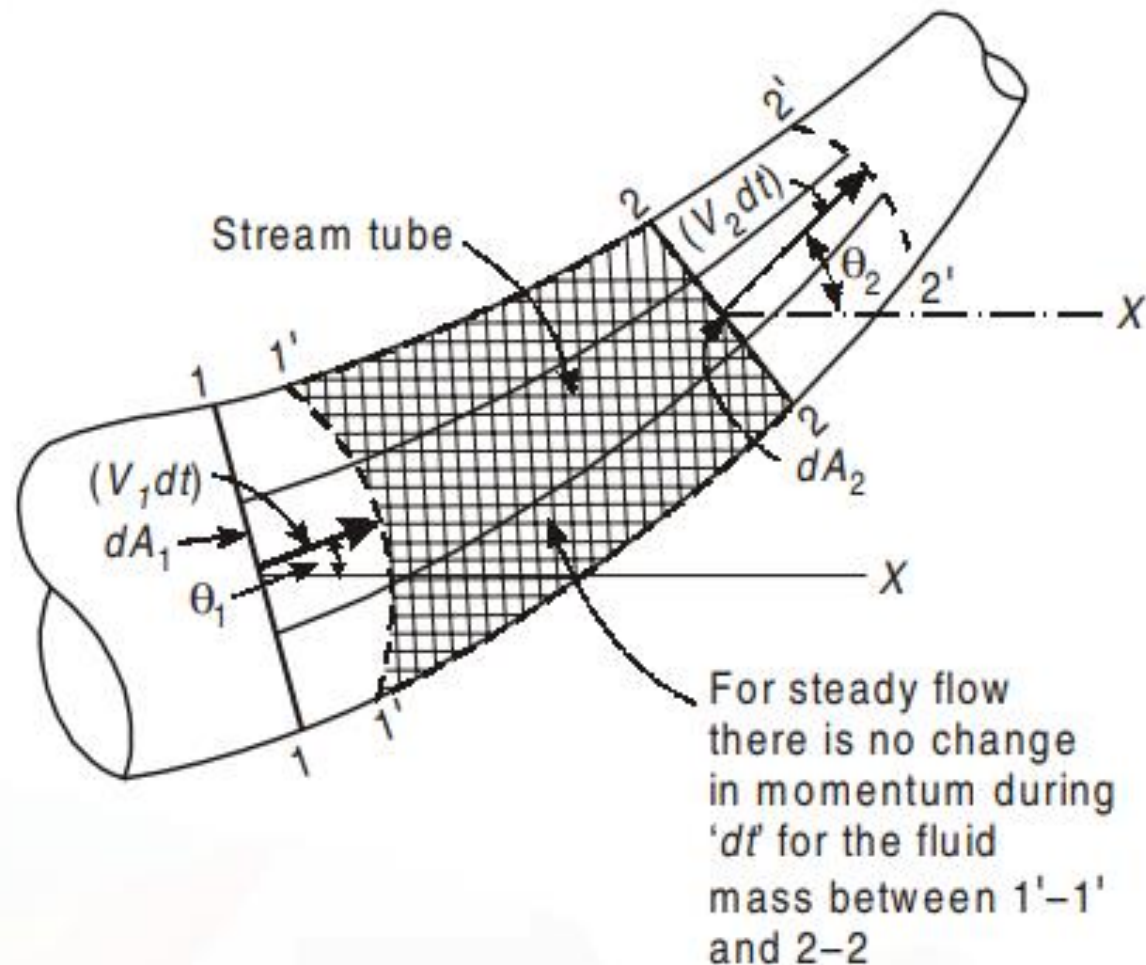
Lecture 25: Fluid Dynamics

Momentum Equation

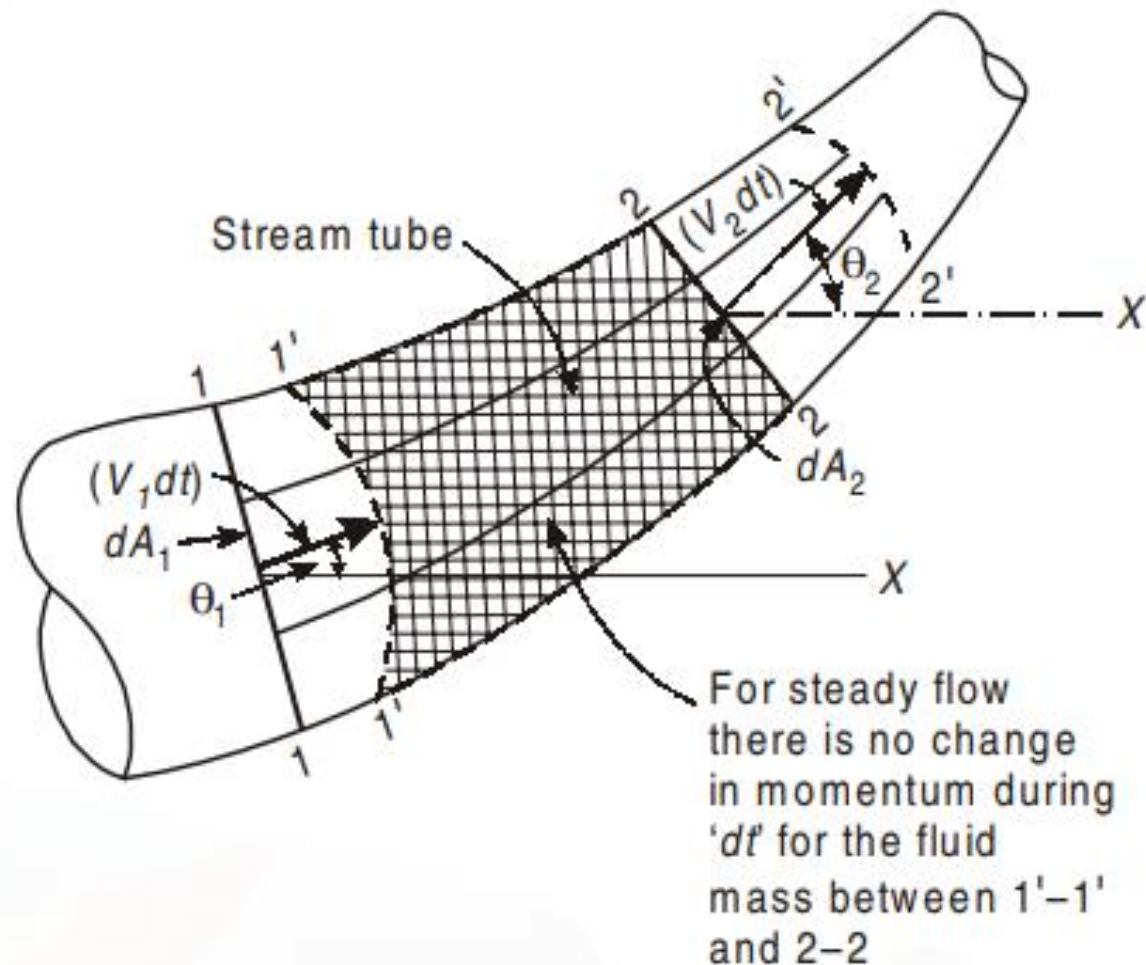
Net force in X (or Y) direction = rate of change in momentum in X (or Y) direction



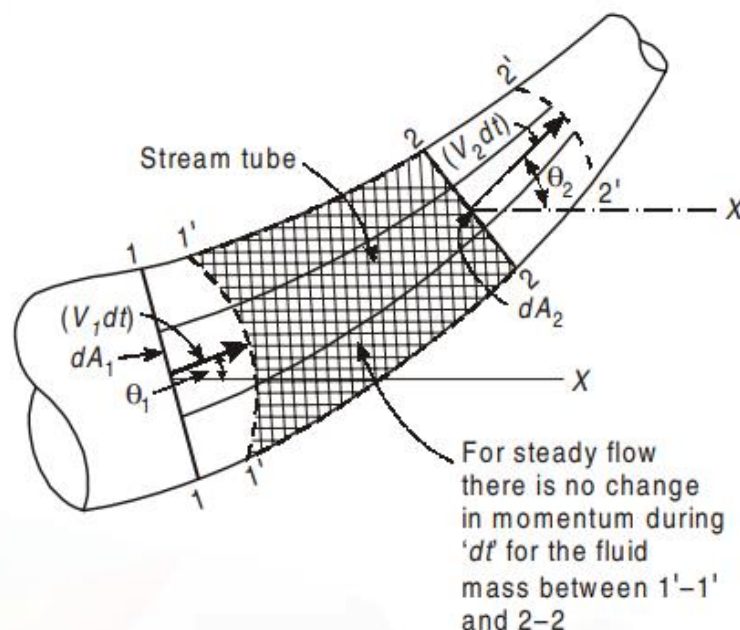
$$\left[\begin{array}{c} \text{Change in momentum} \\ \text{of fluid mass} \\ \text{during } dt \end{array} \right]_x = \left[\begin{array}{c} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1'-1' \text{ and } 2'-2' \end{array} \right]_x - \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 2-2 \end{array} \right]_x$$



$$\left[\begin{array}{l} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1'-1' \text{ and } 2'-2' \end{array} \right]_x = \left[\begin{array}{l} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1'-1' \text{ and } 2-2 \end{array} \right]_x + \left[\begin{array}{l} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 2-2 \text{ and } 2'-2' \end{array} \right]_x$$



$$\left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 2-2 \end{array} \right]_x = \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 1'-1' \end{array} \right]_x + \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1'-1' \text{ and } 2-2 \end{array} \right]_x$$



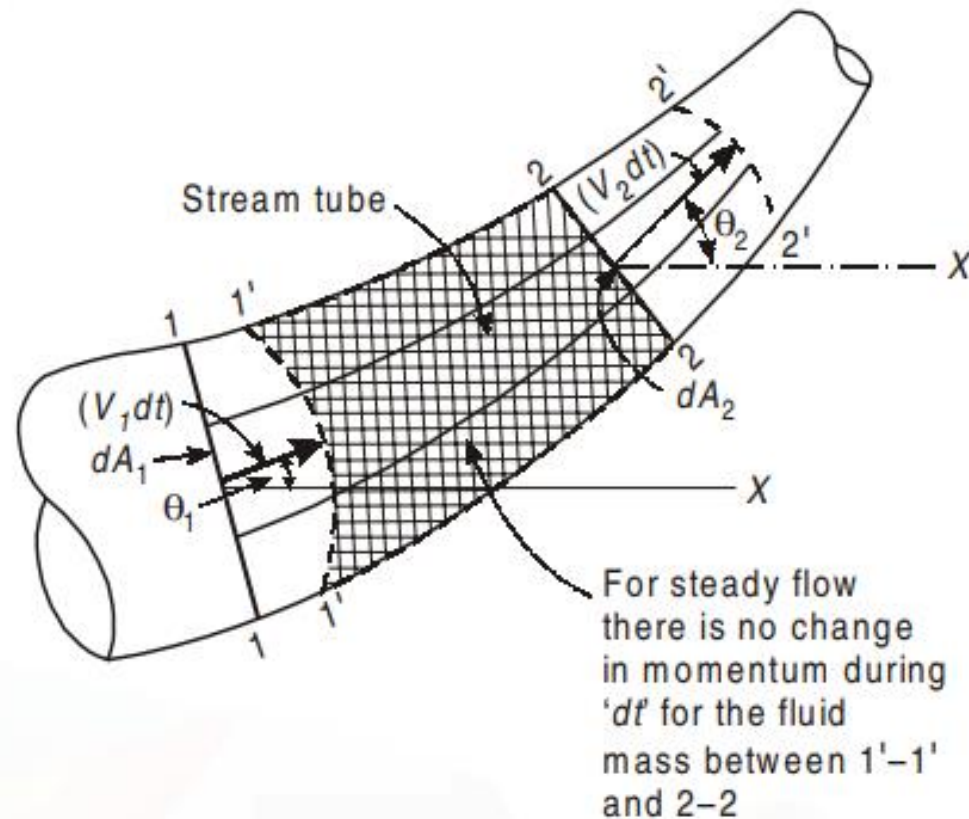
$$\left[\begin{array}{c} \text{Change in momentum} \\ \text{of fluid mass} \\ \text{during } dt \end{array} \right]_x = \left[\begin{array}{c} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1'-1' \text{ and } 2'-2' \end{array} \right]_x - \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 2-2 \end{array} \right]_x$$

But

$$\left[\begin{array}{c} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1'-1' \text{ and } 2'-2' \end{array} \right]_x = \left[\begin{array}{c} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1'-1' \text{ and } 2-2 \end{array} \right]_x + \left[\begin{array}{c} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 2-2 \text{ and } 2'-2' \end{array} \right]_x$$

and

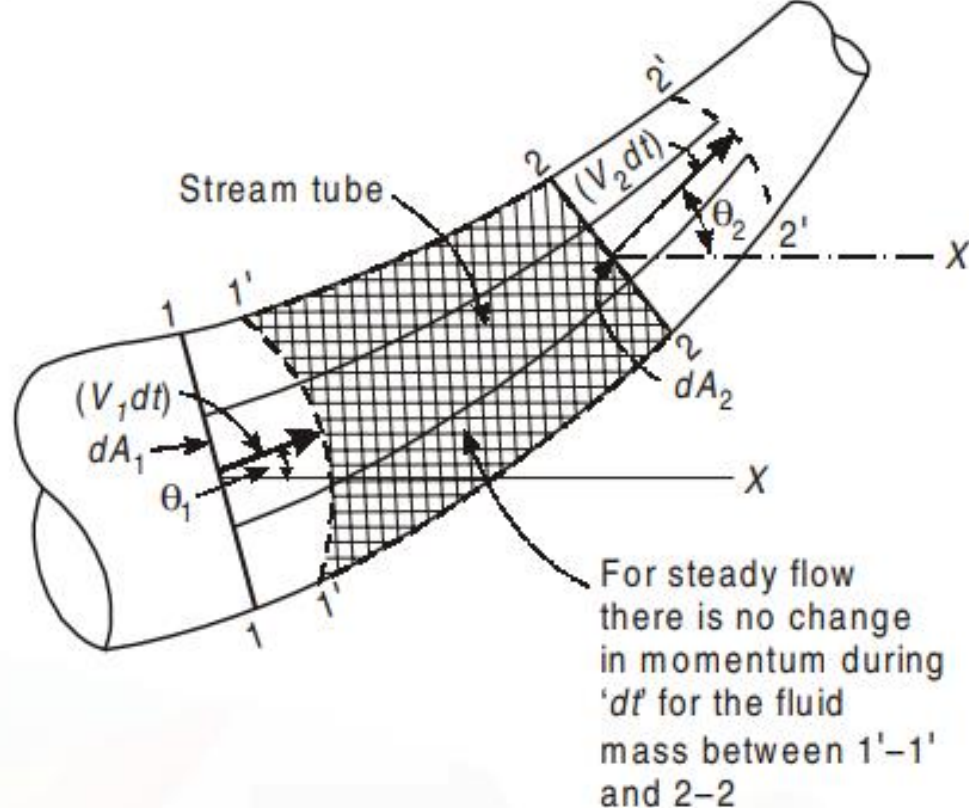
$$\left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 2-2 \end{array} \right]_x = \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 1'-1' \end{array} \right]_x + \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1'-1' \text{ and } 2-2 \end{array} \right]_x$$



$$\left[\begin{array}{l} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 1' - 1' \text{ and } 2 - 2 \end{array} \right]_x = \left[\begin{array}{l} \text{Momentum at} \\ t \text{ of fluid} \\ 1' - 1' \text{ and } 2 - 2 \end{array} \right]_x$$

From which it follows that

$$\left[\begin{array}{l} \text{Change in momentum} \\ \text{of fluid mass} \\ \text{during } dt \end{array} \right]_x = \left[\begin{array}{l} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 2 - 2 \text{ and } 2' - 2' \end{array} \right]_x - \left[\begin{array}{l} \text{Momentum at} \\ t \text{ of fluid} \\ 1 - 1 \text{ and } 1' - 1' \end{array} \right]_x$$



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$$\left[\begin{array}{c} \text{Change in momentum} \\ \text{of fluid mass} \\ \text{during } dt \end{array} \right]_x = \left[\begin{array}{c} \text{Momentum at} \\ (t + dt) \text{ of fluid} \\ 2-2 \text{ and } 2'-2' \end{array} \right]_x - \left[\begin{array}{c} \text{Momentum at} \\ t \text{ of fluid} \\ 1-1 \text{ and } 1'-1' \end{array} \right]_x$$

The above relationship when expressed in terms of mathematical symbols, it becomes

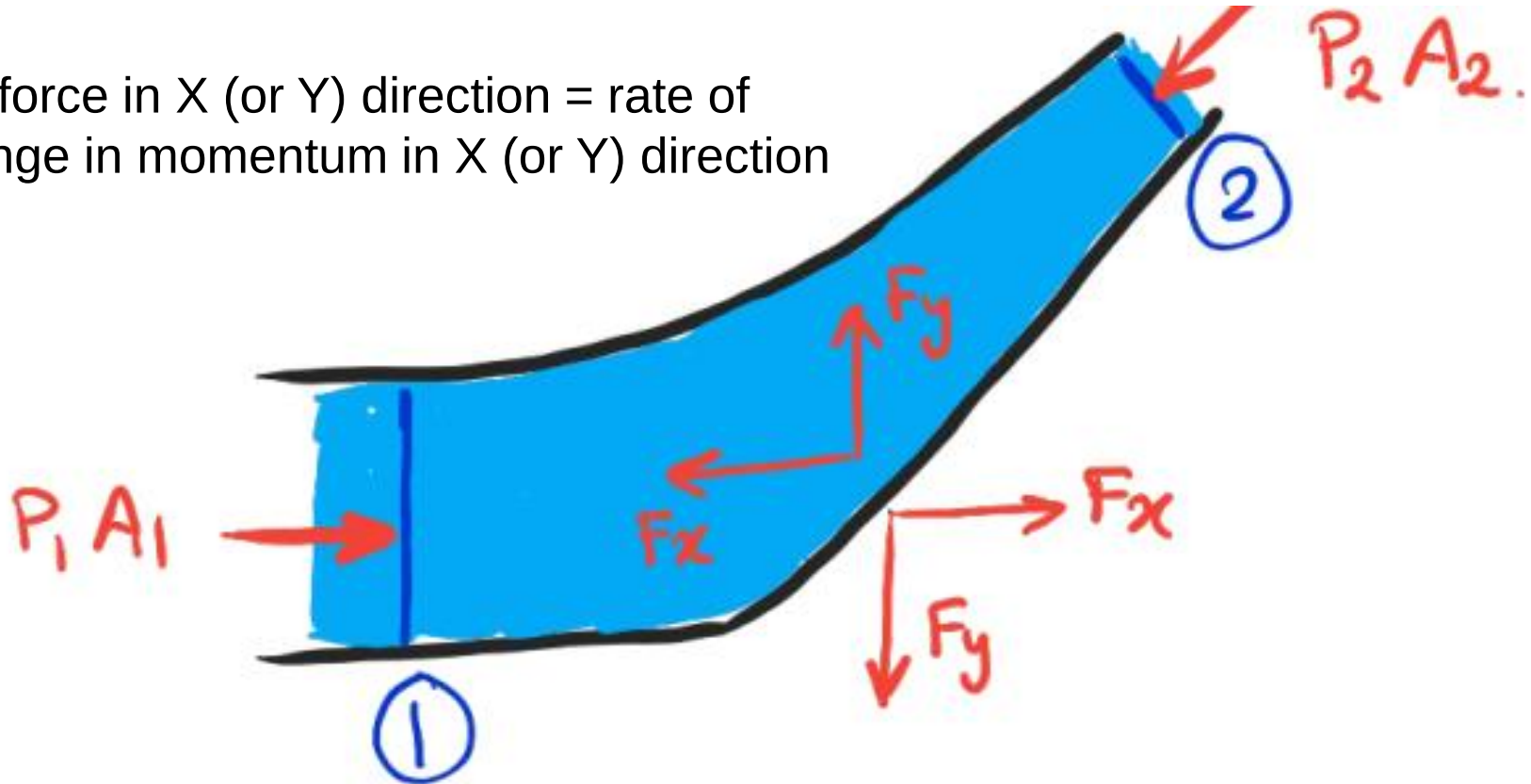
$$\sum d(mv_x) = \int_{A_2} \rho v_2 dt dA_2 (v_2)_x - \int_{A_1} \rho v_1 dt dA_1 (v_1)_x$$

Momentum Equation



A 45° reducing bend is connected in a pipe line, the diameters at the inlet and outlet of the bend being 40 cm and 20 cm respectively. Find the force exerted by water on the bend if the intensity of pressure at inlet of bend is 21.58 N/cm^2 . The rate of flow of water is 500 litres/s. [Ans. 22696.5 N ; $20^\circ 3.5'$]

Net force in X (or Y) direction = rate of change in momentum in X (or Y) direction



Momentum Equation

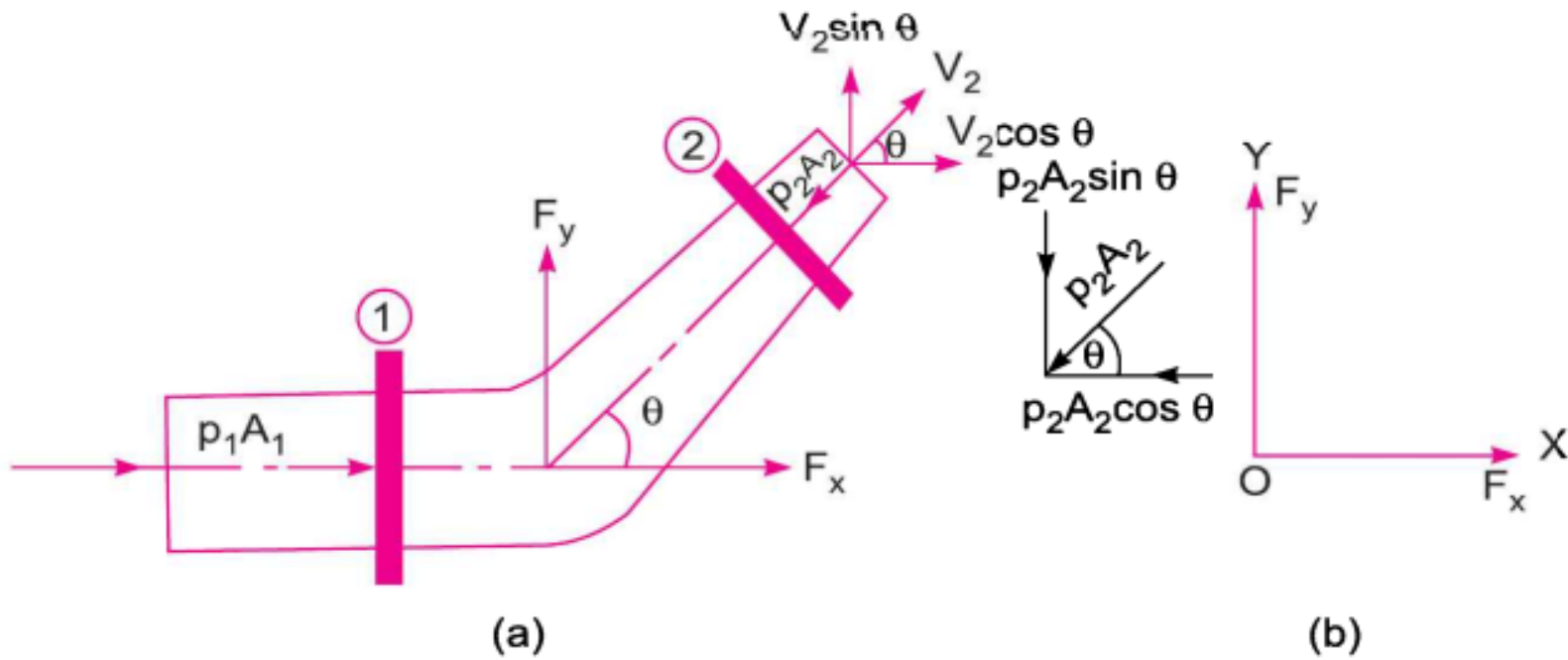


Fig. 6.18 Forces on bend.

Net force in X (or Y) direction = rate of change in momentum in X (or Y) direction

$$F_x = \rho Q (V_1 - V_2 \cos \theta) + p_1 A_1 - p_2 A_2 \cos \theta$$

$$F_y = \rho Q (-V_2 \sin \theta) - p_2 A_2 \sin \theta$$

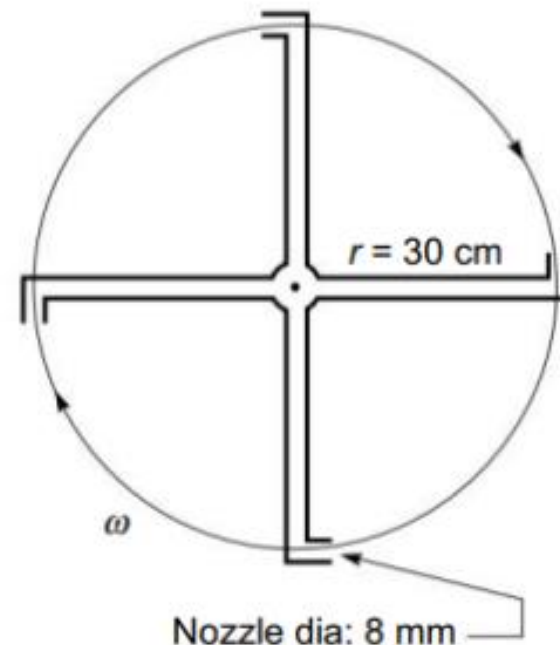
Moment of Momentum Equation

- Required for
 - Analysis of Sprinkler problems
 - Analysis of flow problems in turbines and centrifugal pumps

- Moment of momentum principle – Resulting/net torque is equal to rate of change in moment of momentum

A four-arm lawn sprinkler, shown in Fig. 5.30 has provision for admitting water at the axis of rotation at a rate of 1.5 L/s. calculate the steady rotational speed by neglecting friction.

(Ans. $N = 237.5$ rpm)





THANK YOU